

Table 2-7: Halving/Doubling Table, Part 7

Halving	Doubling
89	18×1
44	18×2
22	18×4
11	18×8
5	18×16
2	18×32
1	18×64

The doubling column is now written in terms of 1, 2, 4, 8, and so on to 64. These are powers of 2, and we can also write them as 2^0 , 2^1 , 2^2 , and so on. When we take our final sum (adding together the doubling rows with odd entries in the halving column), we're really finding this sum:

$$18 \times 2^0 + 18 \times 2^3 + 18 \times 2^4 + 18 \times 2^6 = 18 \times (2^0 + 2^3 + 2^4 + 2^6) = 18 \times 89$$

The fact that RPM works hinges on the fact that

$$(2^0 + 2^3 + 2^4 + 2^6) = 89$$

If you look closely enough at the halving column, you can get a sense for why the preceding equation is true. We can also write this column in terms of powers of 2 ([Table 2-8](#)). When we do so, it's easier to start at the lowest entry and work upward. Remember that 2^0 is 1 and 2^1 is 2. In every row, we multiply by 2^1 , and in the rows where the halving number is odd, we also add 2^0 . You can see the expression start to resemble our equation more and more as you rise through the rows. By the time we reach the top of the table, we have an expression that simplifies to exactly $2^6 + 2^4 + 2^3 + 2^0$.